

At withdrawal, the earnings are taxed as ordinary income. The current limit for non-deductible IRA contributions is \$5,000 for those age forty-nine or younger and \$6,000 for those fifty and older.

Equity Fund: pays capital gains annually. In this case, the earnings are taxed each year at the 15% capital gains rate:

$$1,000 \times (1 + 0.12(1 - 0.238))^n. \quad (12.3)$$

This category of funds includes assets whose prices are marked up to market value at year end, even when no sale has occurred.

Non-Dividend Paying Stock: There is a reason why Warren Buffet has never allowed Berkshire Hathaway to pay a dividend. This investment is similar to the non-deductible IRA, except the earnings are taxed as a capital gain rather than as income:

$$\underbrace{1,000 \times (1 + 0.12)^n}_{\text{Pre-tax accumulation}} - \underbrace{0.238[1,000 \times (1 + 0.12)^n - 1,000]}_{\text{Taxed at withdrawal as capital gain}}. \quad (12.4)$$

Tax Exempt or Tax Deferred: After n years, a tax-exempt investment simply cumulates to

$$1,000 \times (1 + 0.12)^n. \quad (12.5)$$

The formula is exactly the same for a municipal bond (although in practice a municipal bond's average return would be much lower, see Section 3) as for a tax-deductible retirement account. Every dollar contributed to the latter is equivalent to $(1 - 0.396) = \$0.60$ after tax. But, at withdrawal, the pension is taxed as regular income at the full 39.6% rate. Thus we have

$$1,000 \times \frac{1}{(1 - 0.396)} (1 + 0.12)^n (1 - 0.396) = 1,000 \times (1 + 0.12)^n \quad (12.6)$$

as the return on a pension investment, which is exactly the same as equation (12.4). This category also includes 529 college savings plans.

We know immediately that the bond fund is always the worst and that the tax-exempt or tax-deferred fund is always the best (remember, everything in this simple example earns 12% with certainty). This implies that people should not be holding bonds in taxable accounts, unless they are very risk averse and their entire portfolio, both taxable and tax-deferred, is dominated by bonds. What about the other investments?